

Accessibility Test Document

Alt Text for Images via BibLaTeX Metadata

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Abstract

This document demonstrates the use of accessible graphics commands that automatically pull alt text descriptions from BibLaTeX metadata files. The alt text is stored in the **abstract** field of each bibliography entry and is embedded into the PDF for screen reader compatibility.

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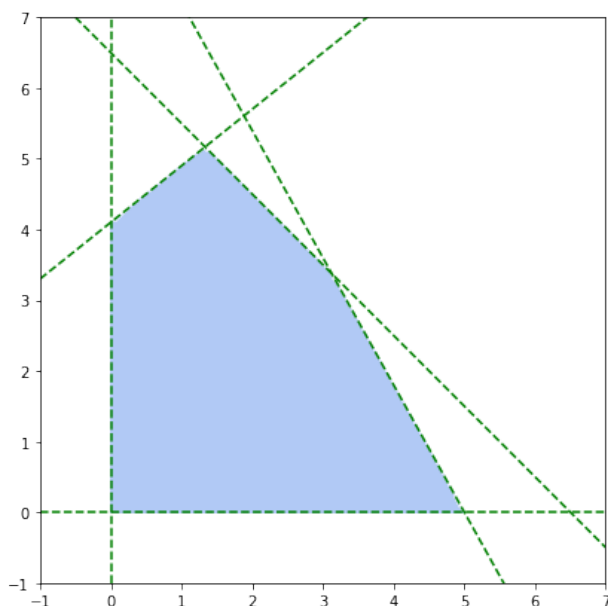
1 Introduction

This document tests the accessible graphics functionality. Each image command automatically:

1. Includes the image from the appropriate figures folder
2. Adds alt text from the **abstract** field in the **.bib** file
3. Displays copyright attribution
4. Adds a footnote with full metadata (title, URL, author, date)

2 Inline Graphics (No Figure Environment)

The `\includegraphicstatic` command includes an image inline (centered, but not in a float). Here is an example showing a linear programming feasible region:



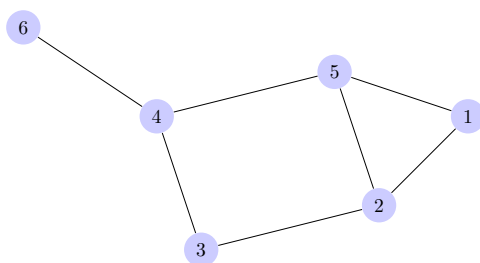
© Robert Hildebrand [CC BY-SA 4.0]¹

¹*Linear Programming Feasible Region*, from . Robert Hildebrand [CC BY-SA 4.0], 2023.

The alt text for this image (stored in the bib file) describes the polygonal feasible region and constraint lines.

3 Inline Graphics from figures-source

The `\includegraphicsource` command includes an image from the figures-source folder. Here is an example showing an undirected graph:



© Robert Hildebrand²

This tests that the alt text and metadata are correctly pulled from the figures-source metadata file.

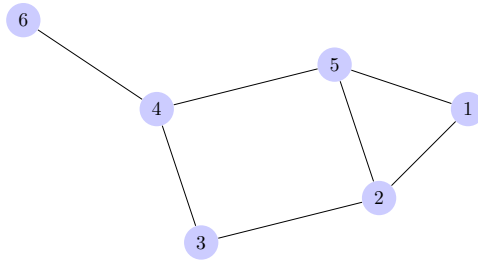
4 No-Cite Versions (No Attribution)

For graphics you created yourself where attribution isn't needed, use the “nocite” versions. These still include alt text for accessibility but omit the copyright line and footnote metadata.

4.1 Inline No-Cite

The `\includegraphicsourcenocite` command:

²*Undirected Graph Example*, from . Robert Hildebrand, 2026.



Notice there's no copyright line or footnote below the image.

4.2 Figure Environment No-Cite

The `\includefiguresourcenocite` command creates a figure with caption but no attribution:

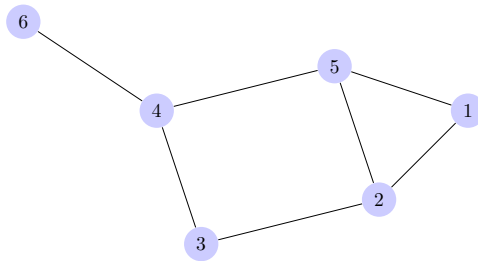


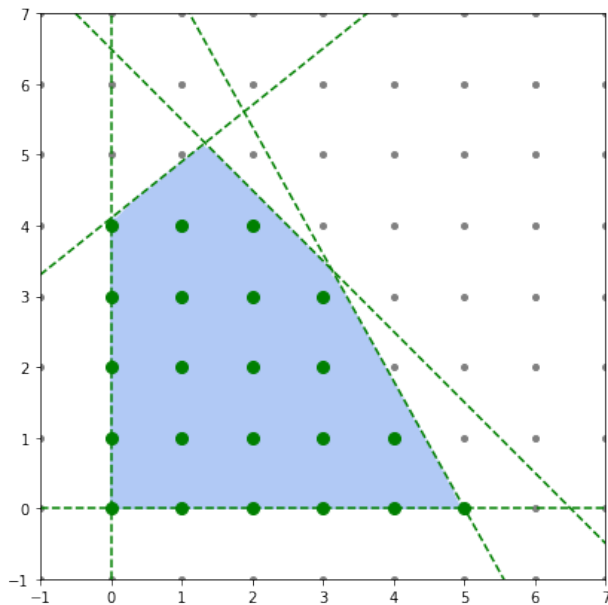
Figure 1: Undirected graph example without attribution

5 Figure Environment with Auto-Caption

The `\includefigurestatic` command creates a proper figure environment with a caption pulled from the bibliography title field.

You can reference this figure using `\autoref{fig:IP-feasible-region.png}`, which produces: Figure 2.

³*Integer Programming Feasible Region*, from . Robert Hildebrand [CC BY-SA 4.0], 2023.



© Robert Hildebrand [CC BY-SA 4.0]³

Figure 2: *Integer Programming Feasible Region*

6 Figure with Custom Caption

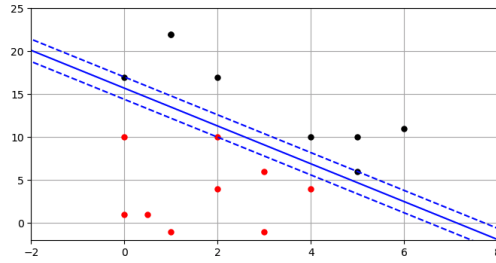
You can also provide a custom caption as the first optional argument:

The alt text still comes from the bib file’s abstract field, but the caption is overridden.

7 Manual Alt Text Override

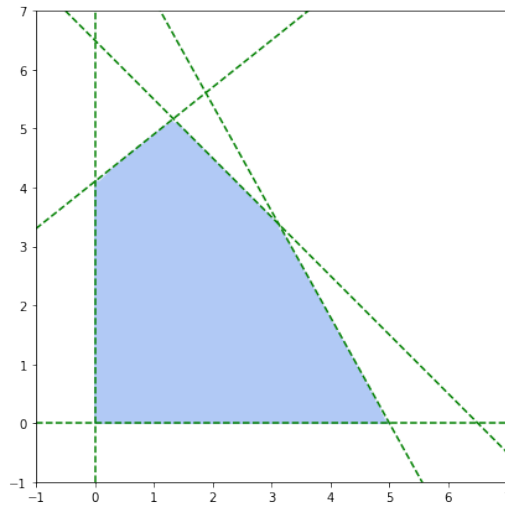
For cases where you need to provide alt text directly (perhaps context-specific), use the `\includegraphicstaticalt` command:

⁴*Support Vector Machine Classification*, from . Robert Hildebrand [CC BY-SA 4.0], 2023.



© Robert Hildebrand [CC BY-SA 4.0]⁴

Figure 3: A support vector machine separating two classes of data



© Robert Hildebrand [CC BY-SA 4.0]⁵

This overrides the alt text from the bib file while keeping all other meta-data.

8 How It Works

8.1 The Metadata File

Alt text is stored in the **abstract** field of each bib entry:

⁵*Linear Programming Feasible Region*, from . Robert Hildebrand [CC BY-SA 4.0], 2023.

```
@Online{LP-feasible-region.png,
  author = {{Robert Hildebrand [CC BY-SA 4.0]}},
  title = {Linear Programming Feasible Region},
  abstract = {A two-dimensional coordinate system
    showing a shaded polygonal region representing
    the feasible region of a linear program...},
  year = {2023},
  options = {skipbib=true},
}
```

8.2 The Document Setup

The key requirement is the `\DocumentMetadata` command *before* `\documentclass`:

```
% For TeX Live 2025 (January release):
\DocumentMetadata{
  lang=en-US,
  pdfversion=2.0,
  testphase={phase-III}
}
\documentclass{article}

% For later releases (2025-11-01+), use:
% \DocumentMetadata{lang=en-US, pdfversion=2.0, tagging=on}
```

8.3 Available Commands

9 Verifying Accessibility

To verify that alt text is correctly embedded:

1. Open the PDF in Adobe Acrobat Pro
2. Go to **View** → **Navigation Panels** → **Tags**
3. Expand the tag tree and find **<Figure>** tags
4. Right-click and select **Properties** to see the alt text

Command	Folder	Float
<code>\includegraphicstatic</code>	figures-static/	No
<code>\includegraphicsource</code>	figures-source/	No
<code>\includefigurestatic</code>	figures-static/	Yes
<code>\includefiguresource</code>	figures-source/	Yes
<code>\refincludefigurestatic</code>	figures-static/	Yes + autoref
<code>\refincludefiguresource</code>	figures-source/	Yes + autoref
<code>\includegraphicstaticalt</code>	figures-static/	No (manual alt)
<code>\includegraphicsourcealt</code>	figures-source/	No (manual alt)

Table 1: Summary of accessible graphics commands

5. Or use **Accessibility** → **Full Check** to run an accessibility audit

Alternatively, use a screen reader (NVDA, JAWS, VoiceOver) to test that images are announced with their descriptions.

References

- [1] Robert Hildebrand. *Undirected Graph Example*. Extracted from part1-linear-programming/ch02-modeling/modeling-sums-continued.tex, line 31. 2026 (cit. on pp. 4, 5).